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ABSTRACT

The development of high thermal conductivity thin film materials for the thermal management of electronics requires accurate and precise methods for characterizing heat spreading capability, namely, in-plane thermal conductivity. However, due to the complex nature of thin film thermal property measurements, resolving the in-plane thermal conductivity of high thermal conductivity anisotropic thin films with high accuracy is particularly challenging. Capable transient techniques exist; however, they usually measure thermal diffusivity and require heat capacity and density to deduce thermal conductivity. Here, we present an explicit uncertainty analysis framework for accurately resolving in-plane thermal conductivity via two independent steady-state thermometry techniques: particle-assisted Raman thermometry and electrical resistance thermometry. Additionally, we establish error-based criteria to determine the limiting experimental conditions that permit the simplifying assumption of one-dimensional thermal conduction to further reduce thermal analysis. We demonstrate the accuracy and precision (<5% uncertainty) of both steady-state techniques through in-plane thermal conductivity measurements of anisotropic nanocrystalline diamond thin films.

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I. INTRODUCTION

The high local power density of radio frequency (RF) and power electronic devices requires innovative thermal management techniques in order to create devices that operate reliability and within thermal design limits. One method to help facilitate these thermal management solutions includes the integration of these devices with high thermal conductivity substrates and thin films such as polycrystalline diamond for lateral heat spreading, which reduces the density of the heat being dissipated.[1,2] In addition to polycrystalline diamond, there exist recent reports of other high thermal conductivity materials[3–7] that show promise as thin film heat spreaders for future generations of wide bandgap and ultra-wide bandgap power and RF devices.[8] For novel anisotropic high thermal conductivity thin film materials such as CVD diamond, hexagonal boron nitride (hBN), and BAs, accurate and precise methods for characterizing in-plane thermal conductivity, $k_{\parallel}$, are critical. However, due to the complexities that are inherent in thin film thermal property measurements, resolving $k_{\parallel}$ with high accuracy and minimal uncertainty is challenging.[9–14]

Despite the practical challenges of sample preparation, experimental precision, and complex analysis, several measurement techniques have been demonstrated to characterize $k_{\parallel}$ of thin films (<3 μm).[10,15–20] The most prominent of these techniques, namely, time-domain thermoreflectance (TDTR),[10,16,21] frequency-domain thermoreflectance (FDTR),[19,21] and 3-omega (3ω),[17,18,22] require precise electrical and optical configurations and multi-parameter fitting to an analytical heat transfer solution for the measured sample structure. Alternative to these complex transient techniques, steady-state techniques exist for measuring $k_{\parallel}$ of thin films including Raman thermometry,[15,23–25] electrical resistance thermometry (ERT),[20] and steady-state thermoreflectance (SSTR).[26] The primary

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advantages of steady-state techniques over transient alternatives for characterizing $k_{\parallel}$ of thin films include (i) a simplified experimental setup and thermal analysis and (ii) eliminating the need for heat capacity and density to deduce $k_{\parallel}$. However, despite these practical advantages, steady-state techniques are particularly prone to high measurement uncertainty and error due to parasitic heat losses.[20] Failure to rigorously consider the experimental sources of uncertainty and validity of model assumptions leads to inaccurate $k_{\parallel}$ measurements as shown by numerical simulations of Raman thermometry-based thermal conductivity measurements.[23] For example, a modified implementation[15] of such steady-state Raman thermometry-based thermal conductivity measurements has demonstrated substantial disagreement between reported $k_{\parallel}$ values measured from near-identical nanocrystalline diamond (NCD) suspended thin films.[24,27,28] The recently developed SSTR technique[26] is well-suited for measuring the thermal conductivity of bulk substrates and has been proposed to be capable of measuring $k_{\parallel}$ of thin films. However, properly resolving $k_{\parallel}$ involves a thorough understanding of the experimental setup, sample design, and sensitivity analysis. Fortunately, more simply implemented steady-state techniques such as particle-assisted Raman thermometry (PaRT)[15,28] and ERT[20] exist. However, an explicit framework for rigorously considering the experimental uncertainty and error associated with measuring $k_{\parallel}$ of thin films via simply implemented steady-state thermometry techniques has not yet been published in the literature.

Therefore, to demonstrate the efficacy and practical limitations of using steady-state thermometry techniques to characterize $k_{\parallel}$ of thin film heat spreading materials, we present an explicit uncertainty analysis framework for accurately resolving $k_{\parallel}$ via two independent steady-state thermometry techniques: PaRT[29] and ERT.[20] Furthermore, we have established error-based criteria to determine the limiting experimental conditions that permit the simplifying assumption of 1D thermal conduction to drastically reduce the post-measurement analysis necessary for deducing $k_{\parallel}$ with minimal sacrifice to measurement accuracy. We have validated the accuracy and precision of our uncertainty analysis methodology with experimental $k_{\parallel}$ results measured from anisotropic NCD thin films.

II. SAMPLE PREPARATION AND DESIGN

The NCD thin films examined in this work were grown to thicknesses of 1 and 3 μm on 75-mm-diameter and 200-μm-thick silicon wafers via microwave plasma-enhanced chemical vapor deposition. Prior to diamond growth, the silicon substrates were seeded with nanocrystalline diamond powders for NCD nucleation. Each diamond-on-silicon wafer was diced to produce an array of 10×10 mm^2 square dies that were processed into suspended membrane samples by removing the silicon substrate in selected areas using plasma-based etching. A total of six samples were examined: two from wafer 1 (labeled 1a and 1b), two from wafer 2 (labeled 2a and 2b), one from wafer 3, and one from wafer 4. The samples taken from wafers 1 and 2 were grown from NCD nucleation seed sizes of 4 and 20 nm, respectively, to a thickness of 1 μm under similar growth conditions. The samples taken from wafers 3 and 4 were grown to thicknesses of 1 and 3 μm, respectively, under similar growth conditions. To create a controllable heat source and enable electrical measurements, 5-μm-wide metal lines and four-point probe contact pads were patterned onto the NCD films by depositing a 10-nm-thick titanium adhesion layer followed by 200 nm of gold. As shown in Fig. 1, the center metal line was used as a resistive heater (Joule heating) to dissipate heat across the membrane width. Additionally, an ~100-nm-thick thin film of aluminum (Al) was deposited on the backside of samples 1a, 1b, 2a, 2b, and 4 as a by-product of processing necessary for other samples on each source wafer. Due to the high thermal conductivity of the NCD and Al films, a negligible temperature gradient was observed from our finite element model (FEM) (detailed in the supplementary material) in the through-thickness direction of the films. For the in-plane direction, the lateral heat conduction is two-dimensional (2D) in nature. However, for some aspect ratios, the approximation of one-dimensional (1D) conduction can be used to model the thermal response of the membrane. A parametric study performed using finite element analysis demonstrated that 1D conduction could be assumed in the center of the suspended membrane for the length-to-width aspect ratio of 5:1 (see Sec. V). Thus, all our samples were designed to this criterion with suspended membrane dimensions of 400×2000 μm^2. The center portion of the suspended membrane, where 1D conduction can be assumed, is hereafter referred to as the 1D conduction regime. The criteria for this 1D conduction assumption are discussed in further detail in Sec. V. A schematic of the suspended membrane samples examined is shown in Fig. 1.

III. OVERVIEW OF STEADY-STATE TECHNIQUES

Two independent steady-state thermometry techniques, PaRT and ERT, were used to measure $k_{\parallel}$ of the six NCD suspended membrane samples fabricated according to two types of growth conditions. The in-plane thermal conductivity measurement via both steady-state thermometry techniques was accomplished according

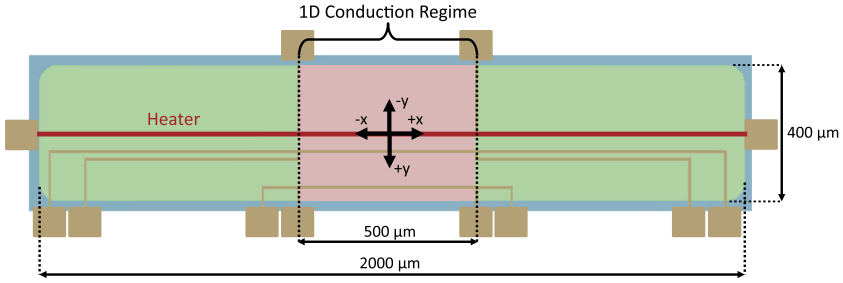


FIG. 1. Schematic of a suspended membrane sample. The center metal line was used as a resistive heater to dissipate heat across the membrane. The 1D conduction regime is highlighted for clarity.

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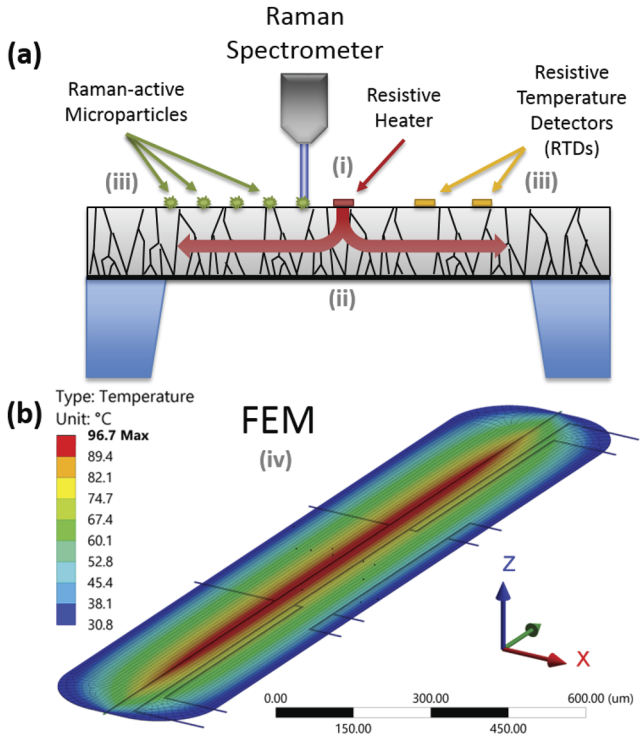


FIG. 2. Schematic of the PaRT and ERT (a) experiments and (b) the 3D steady-state thermal FEM. The $k_{\parallel}$ measurement procedure included (i) introducing a steady-state heat source, (ii) physically confining thermal transport to the in-plane dimension, (iii) measuring the imposed temperature gradient, and (iv) fitting the experimental results to a numerical FEM.

to the following general procedure as illustrated in Fig. 2: (i) introduce a steady-state heat source, (ii) physically confine thermal transport to the in-plane dimension, (iii) measure the imposed temperature gradient, and (iv) fit the experimental temperature results to a theoretical model to determine the best fit $k_{\parallel}$. The theoretical model used to deduce $k_{\parallel}$ was a 3D steady-state thermal FEM implemented in ANSYS Workbench. Additional details about the FEM including thermal properties and boundary conditions are described in the supplementary material.

A. Particle-assisted Raman thermometry

Micro-Raman thermometry was used to measure the lateral temperature gradient across the NCD suspended membrane samples when constant electrical power was supplied to the centerline resistive heater. Beechem and Serrano[30] present a comprehensive summary of the methods commonly used to perform micro-Raman thermometry, i.e., correlate the Raman spectrum with lattice temperature, and compare the advantages and challenges inherent to each method. We have implemented the peak position method of micro-Raman thermometry because of its linear correlation between the Raman peak position shift and temperature rise. However, despite the experimental practicality of this method, the Raman peak position, ω, is a measure of the frequency of lattice vibrations and is, therefore, particularly sensitive to nonthermal biasing. For example, as a material experiences local or self-heating, the interatomic forces are typically altered by a combination of effects: (i) volumetric expansion that accompanies the increasing lattice temperature and (ii) thermoelastically induced strain. Since the frequency of lattice vibrations indiscriminately depends on the interatomic forces, a measure of peak position shift with respect to a reference condition, $\omega - \omega_o$, contains a combination of temperature and thermomechanical stress information. According to the conclusions of numerical error analysis presented in the literature,[23] thermomechanical-induced errors in Raman temperature measurements can invalidate thermal conductivity deductions through error propagation.

However, due to the material-specific nature of Raman thermometry, it is possible to altogether avoid thermomechanical signal biasing by probing the temperature of nanoparticles that are deposited on the surface and have a distinct Raman signature that is different from the underlying suspended membrane. Raman thermometry of surface-deposited temperature sensors has been demonstrated in the literature using a variety of temperature sensor materials including diamond microparticles,[31] silicon nanowires,[15] titanium dioxide (TiO_2) nanoparticles,[29] and hexagonal boron nitride (hBN) nanoparticles.[28] To identify the most suitable temperature sensor, it is necessary to consider the peak positions and relative intensities of the Raman spectra of the temperature sensor and the suspended membrane to avoid peak superposition and a poor signal-to-noise ratio. Consequently, diamond microparticles and silicon nanowires were inappropriate temperature sensors for our experiments characterizing NCD grown on silicon substrates due to Raman peak superposition. Furthermore, as illustrated in Fig. 3, hBN was also ill-suited for our experiments due to the poor signal-to-noise ratio caused by the large diamond luminescence background concomitant with our visible laser excitation (488 nm). Although not implemented in this work, this poor signal-to-noise ratio of hBN particles on NCD can be resolved by performing Raman thermometry using a near-UV excitation laser.[28] Therefore, based on the distinct Raman signature shown in Fig. 3, TiO_2 particles were used as surface-deposited temperature sensors for micro-Raman thermometry in this work. The TiO_2 particles were deposited onto

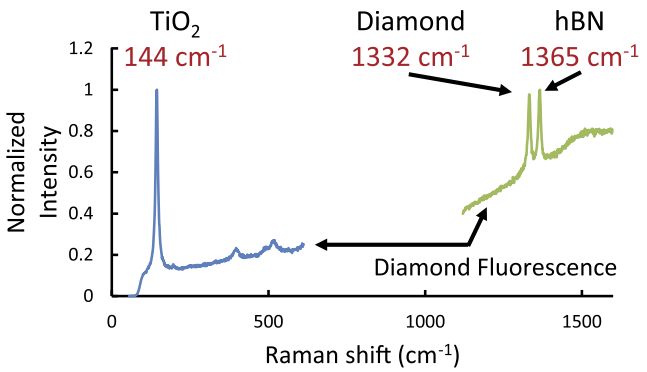


FIG. 3. Measured Raman spectra of the TiO_2 and hBN particles deposited on the NCD suspended membranes. Due to the diamond fluorescence background present in the Raman spectra, TiO_2 was selected as the most suitable surface-deposited temperature sensors for Raman thermometry.

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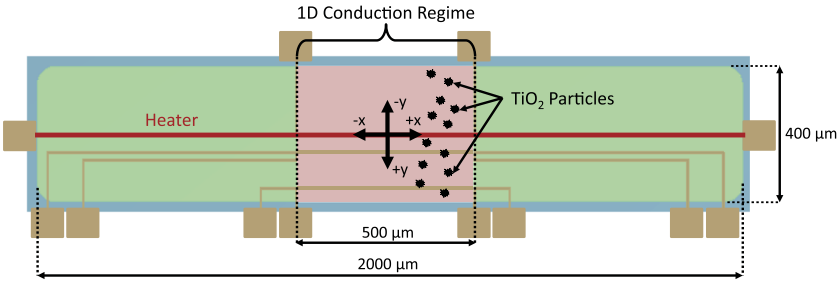


FIG. 4. Schematic of the suspended membrane sample with TiO_2 particles for PaRT. TiO_2 particles located within the 1D conduction regime were selected as local temperature sensors.

the NCD suspended membrane samples by drop-casting a suspension of a TiO_2 nanopowder (anatase phase, 99.9% purity, and 32 nm average particle size) in isopropyl alcohol while maintaining a sample temperature of 85 °C (just above the boiling point of isopropyl alcohol) to accelerate evaporation.

As illustrated in Fig. 4, the surface-deposited TiO_2 nanoparticle aggregates (referred to as particles for simplicity) with a diameter of 1–2 μm and located within the 1D conduction regime were used as local temperature sensors for measuring the temperature distribution across the NCD suspended membranes. A Keithley 2400 source meter was used to supply electrical power to the centerline heater via a four-point probe technique. The micro-Raman thermometry measurements were collected using a Renishaw InVia Raman system equipped with a 488 nm Ar^+ excitation laser that was focused through a Leica 0.75 NA 100× objective to resolve a 0.79 μm theoretical diffraction limited laser focused spot size.

The temperature dependence of the strongest Raman-active TiO_2 phonon mode frequency at 143 cm^{-1}[32] was calibrated by measuring the Raman peak shift of several TiO_2 particles at a series of sample base temperature increments. For the temperature range of 30–150 °C, the linear TiO_2 calibration coefficient, A_{Ti}, was measured to be 0.0251 ± 0.0009 $cm^{-1}/°C$. This result is in precise agreement with the literature value of 0.0245 ± 0.0008 $cm^{-1}/°C$ measured from anatase TiO_2 nanocrystals.[32] Additional details about this calibration procedure are provided in the supplementary material.

The PaRT experiment for determining $k_\parallel$ of each suspended membrane sample was performed by holding the thermal stage at a 30 °C reference condition, sourcing electrical power to the centerline heater, and collecting micro-Raman thermometry measurements from TiO_2 particles spanning the membrane width (10–20 particles). The PaRT-derived temperature measured from each TiO_2 particle, T_{Ti}, was calculated by the following equation:

$$T_{Ti} = \frac{\omega - \omega_o}{A_{Ti}} + T_o, \tag{1}$$

where A_{Ti} is the measured calibration coefficient, ω and ω_o are the Raman peak positions measured during the power condition (heater ON) and reference condition (heater OFF), respectively, and T_o is the TiO_2 particle temperature during the reference condition (heater OFF). For clarity, the TiO_2 particle temperature during the reference condition (heater OFF) was assumed to be equal to the thermal stage surface temperature and thereby measured via the thermocouple mounted on the surface of the thermal stage (shown in Fig. S1 in the supplementary material). Each T_{Ti} was composed of 60 Raman spectrum acquisitions: 20 acquisitions at the reference condition (heater OFF) *before* the power condition, 20 acquisitions *during* the power condition (heater ON), and 20 acquisitions at the reference condition (heater OFF) *after* the power condition. All of the acquired spectra were fit to a pseudo-Voigt function (linear combination of Gaussian–Lorentzian functions) to extract the peak position, intensity, and full-width half-maximum of each spectrum peak. In addition to the T_{Ti} measurements, a Raman thermometry measurement was collected from the silicon substrate, T_{Si}, on either side of the suspended membrane by measuring the linear calibration coefficient of the silicon substrate, A_{Si}, and otherwise following the same experimental procedure that was used for measuring T_{Ti}. The direct measurement of T_{Si} via the Raman-active silicon phonon mode (~522 cm^{-1})[33] was possible because the excitation laser used (488 nm) transmits through the TiO_2 particles and NCD membrane due to their wide bandgaps[34,35] and is absorbed by the silicon substrate with a shallow penetration depth of ~0.5 μm.[36] These auxiliary T_{Si} measurements were useful for detecting the local temperature rise in the silicon substrate during the power condition (heater ON) that were used to properly determine the constant temperature (Dirichlet) boundary condition used in the FEM.

B. Electrical resistance thermometry

ERT is a commonly used electro-thermal technique for measuring the temperature of an electrically conductive feature by probing the change in its electrical resistance with respect to a reference condition. This technique is capable of low uncertainty and high-resolution temperature measurements when the resistance temperature detector (RTD) is a metal with a linear resistance–temperature correlation within the temperature range of interest. Hence, ERT techniques are particularly useful when a metallic feature is used as a resistive heater (Joule heating) because the heater can be simultaneously used as an RTD. A well-established transient implementation of ERT that leverages this heater/sensor duality to measure the thermal conductivity of thin film materials is the 3ω technique.[22] While the differential 3ω technique is appropriate for measuring the low isotropic thermal conductivity (k < 1 W/m K) of dielectric thin films, resolving the high in-plane thermal conductivity ($k_\parallel$ > 50 W/m K) of a thin film can result in high uncertainty ($\delta k_\parallel$ > 25%) and requires careful consideration of limiting assumptions.[17] Therefore, to enable accurate characterization of our high thermal conductivity ($k_\parallel$ > 50 W/m K) NCD films, a steady-state implementation of ERT is demonstrated in this work.

Similar to the PaRT technique, ERT was used to measure the temperature gradient across the NCD suspended membrane samples when constant electrical power was supplied to the centerline heater. As shown in Fig. 5, the centerline heater (RTD 1) and two

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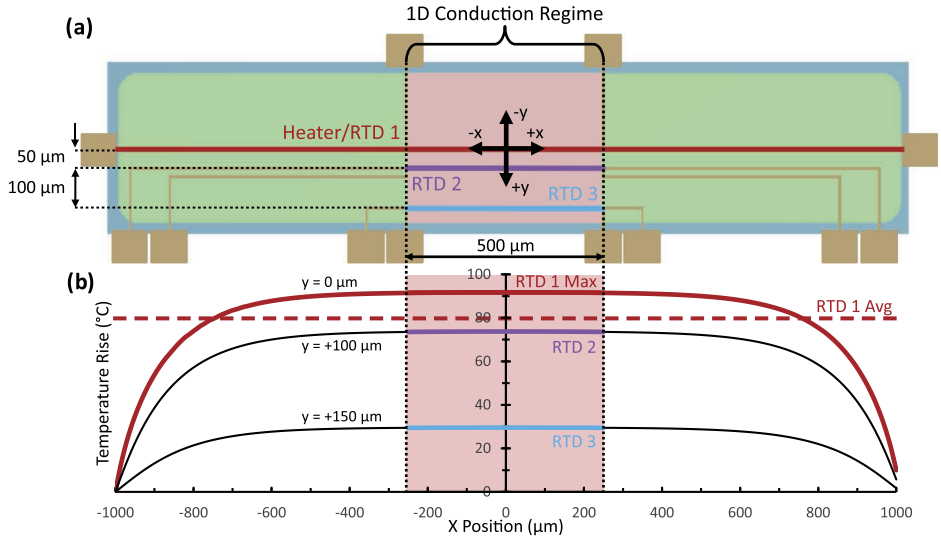


FIG. 5. (a) Schematic of a suspended membrane sample with RTDs for ERT. The 1D conduction regime is highlighted for clarity. (b) Temperature rise across the sample calculated by finite element analysis. The temperature rise measured from each RTD represents a volume-averaged value. Since RTD 2 and RTD 3 are within the 1D conduction regime, each measured temperature is equal to the RTD maximum temperature. However, since RTD 1 spans the entire suspended membrane, the measured temperature of RTD 1 (dashed) is significantly lower than the RTD 1 maximum temperature.

off-center RTDs positioned at offsets of 50 μm (RTD 2) and 150 μm (RTD 3) on the surface of each sample were used to accommodate three independent temperature measurements. A Keithley 2400 source meter was used to supply constant electrical power to and measure the resistance of RTD 1, and an Agilent 34410A 6 ½ Digit Multimeter was used to measure the resistances of RTDs 2 and 3. All the electrical measurements were collected via a four-point probe technique (to mitigate contact resistances) on a Cascade Microtech Summit 11 000 probe station. It is important to note that the temperature rise measured from each RTD represents a volume-averaged value. Since RTDs 2 and 3 are located within the 1D conduction regime, each measured temperature is equal to the RTD maximum temperature. However, since RTD 1 spans the entire suspended membrane, the measured temperature of RTD 1 is significantly lower than the RTD 1 maximum temperature when the sample is locally heated (heater ON). This effect is illustrated in Fig. 5. To more accurately measure the maximum temperature of RTD 1, an additional four-point probe measurement could be used to measure the resistance of only the center of RTD 1.

The procedure for measuring the linear RTD calibration coefficient, A_{RTD}, for each RTD was very similar to the procedure for measuring A_{Ti} and is described in the supplementary material. The ERT experiment for determining $k_{\parallel}$ of each suspended membrane sample was performed by holding the thermal stage at a 30 °C reference condition, sourcing electrical power to the centerline heater and measuring the resistance of each RTD. The ERT-derived temperature measured from each RTD, T_{RTD}, was calculated by the following equation:

$$T_{RTD} = \frac{\frac{R-R_o}{R_o}}{A_{RTD}} + T_o, \tag{2}$$

where, analogous to PaRT, A_{RTD} is the measured calibration coefficient, R and R_o are the resistances measured during the power condition (heater ON) and reference condition (heater OFF), respectively, and T_o is the RTD temperature during the reference condition (heater OFF).

The ERT experiment for determining $k_{\parallel}$ required dramatically less time than the equivalent PaRT experiment. Electrical measurements take less than 1 s, whereas Raman spectrum acquisitions typically require between 1 and 20 s of acquisition time per spectral acquisition. Moreover, each ERT experiment required a total of nine resistance measurements, and each PaRT experiment required a total of 600–1200 Raman spectrum acquisitions. An additional benefit of the time-efficient ERT technique is the ability to quickly approximate the temperature dependence of $k_{\parallel}$ by repeating ERT experiments for various power conditions. By extracting the average temperature of the entire suspended membrane, T_{avg}, for each power condition from the FEM, the temperature-dependent thermal conductivity, $k_{\parallel}(T_{avg})$, of sample 1b was determined for membrane-averaged temperatures of 35–70 °C, as discussed further in Sec. V.

IV. UNCERTAINTY ANALYSIS

To ensure a rigorous estimation of uncertainty in thermal conductivity, we have considered uncertainty contributions from (i) the accuracy, precision, and resolution of our empirical temperature measurements; and (ii) uncertainties of all auxiliary thermal and material properties determined for input to the representative FEM. For clarity, accuracy is defined as a measure of how close the measured value of a property is to the actual value of that property, and precision is defined as a measure of the repeatability of a measured property value. We have propagated each of these primary sources of uncertainty to estimate the total in-plane thermal conductivity uncertainty by a combination of three uncertainty propagation methods: (i) Monte Carlo (MC) simulation, (ii) analytical derivation, and (iii) sensitivity analysis. As illustrated in Fig. 6, analytical derivation was used to quantify temperature measurement uncertainties that were then propagated into an empirical in-plane thermal conductivity uncertainty estimation via Monte Carlo simulation. Sensitivity analysis of the FEM was performed to propagate the auxiliary FEM input parameter (geometry variables, interface conductances, etc.) uncertainties into a computational

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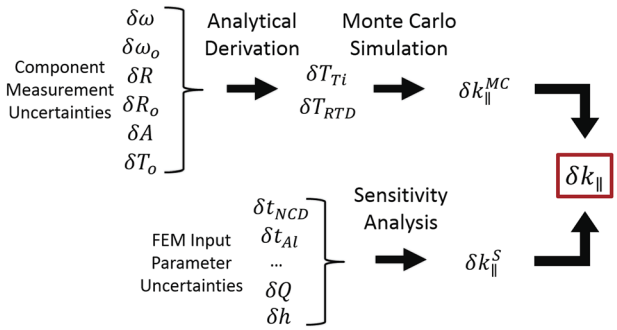


FIG. 6. Flow chart summary of the uncertainty estimation framework. Component measurement uncertainties were propagated into an empirical uncertainty ($\delta k_{\parallel}^{MC}$) via analytical derivation and Monte Carlo simulation. FEM input parameter uncertainties were propagated into a computational uncertainty ($\delta k_{\parallel}^{S}$) via sensitivity analysis. The total in-plane thermal conductivity uncertainty ($\delta k_{\parallel}$) was determined by combining the empirical and computational uncertainties via summation in quadrature [see Eq. (3)].

in-plane thermal conductivity uncertainty estimation. The overall in-plane thermal conductivity uncertainty, $\delta k_{\parallel}$, was estimated by combining the empirical and computational uncertainties via summation in quadrature according to the following equation:

$$\delta k_{\parallel} = \sqrt{\left(\delta k_{\parallel}^{MC}\right)^2 + \left(\delta k_{\parallel}^{S}\right)^2}, \tag{3}$$

where $\delta k_{\parallel}^{MC}$ is the empirical Monte Carlo-based uncertainty in $k_{\parallel}$ and $\delta k_{\parallel}^{S}$ is the computational sensitivity-based uncertainty in $k_{\parallel}$. Furthermore, we have established error-based criteria to determine the limiting experimental conditions that permit the simplifying assumption of 1D thermal conduction to drastically reduce the post-measurement analysis necessary for deducing $k_{\parallel}$.

A. Monte Carlo simulation

In order to deduce an empirical $k_{\parallel}$ for the NCD suspended membrane, we must determine the theoretical (FEM input) NCD $k_{\parallel}$ that yields a computed temperature distribution that best fits our experimental temperature data. We accomplished this parametric optimization through a Monte Carlo simulation executed in MATLAB. Monte Carlo (MC) simulation is an iterative stochastic method for solving optimization problems that is particularly useful for estimating the uncertainty of one or more unknown fitting parameters without knowing how experimental uncertainties will propagate nor how individual input uncertainties may be intercorrelated.[37] Although the full utility of MC simulation can be realized for solving multi-parameter optimization problems that fit experimental data to an analytical model such as those arising from thermoreflectance measurements,[21] MC simulation is still advantageous for our single parameter fitting of $k_{\parallel}$ because there exists no analytical means of propagating experimental temperature measurement uncertainties through an inherently numerical FEM. As illustrated in Fig. 7, our MC simulation propagated experimental temperature uncertainties into a MC-based uncertainty of $k_{\parallel}$, denoted as $\delta k_{\parallel}^{MC}$, by assuming that each experimental temperature measurement was normally

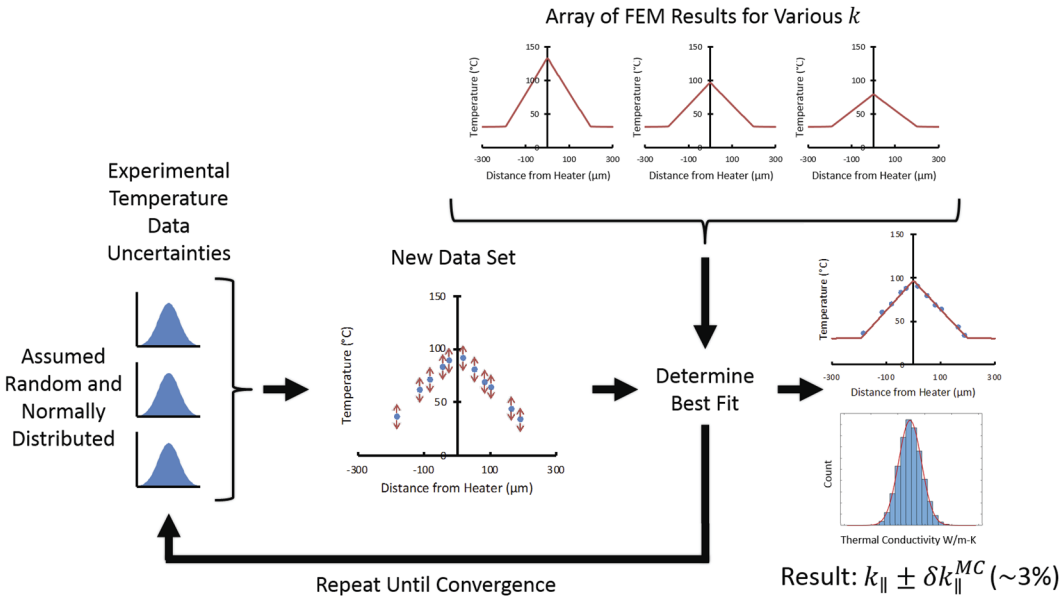


FIG. 7. Flow chart summary of the MC simulation implemented for determining the best-fit $k_{\parallel}$ and $\delta k_{\parallel}^{MC}$. Stochastic experimental temperature uncertainties, δT_{exp}, were propagated by assuming that each T_{exp} was normally distributed about its measured value according to its δT_{exp} and repeatedly fitting randomly sampled experimental datasets to FEM temperature solutions. A normal distribution was fit to the histogram of fitted $k_{\parallel}$ results. The mean value and 95% confidence interval of the $k_{\parallel}$ fit results were evaluated as the best-fit $k_{\parallel}$ and $\delta k_{\parallel}^{MC}$, respectively. Experimental results shown in Table II demonstrate that $\delta k_{\parallel}^{MC} \approx 3\%$ on average.

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distributed about its measured value according to its uncertainty, δT, and repeatedly fitting randomly sampled experimental datasets to FEM temperature solutions. The histogram of fitted $k_{\parallel}$ results was then fit to a normal distribution from which the mean and 95% confidence interval were extracted to be the best-fit $k_{\parallel}$ and $\delta k_{\parallel}^{MC}$, respectively.

B. Analytical derivation

The experimental temperature uncertainty, δT_{exp} (δT_{Ti} or δT_{RTD}), of each thermometry measurement that was input to the MC simulation for determining $k_{\parallel}$ was resolved by analytically propagating the component measurement uncertainties attributed to the limited accuracy, precision, and resolution inherent to each thermometry technique according to the following equation:

$$\delta T_{exp} = \sqrt{\sum_{i=1}^{n}\left(\frac{\partial T}{\partial p_i}\delta p_i\right)^2}, \tag{4}$$

where the partial derivatives were evaluated by differentiating Eqs. (1) and (2) with respect to the n input parameters, p_i, and δp_i is the component measurement uncertainty of parameter, p_i. The propagated δT_{exp} results derived from the PaRT and ERT experiments for evaluating $k_{\parallel}$ of sample 2a are shown in Fig. 8. The measurement-specific procedures that were used to quantify each δp_i are described in Sec. VIII C. Comparison of the relative temperature uncertainties, $\delta T_{exp}/T_{exp}$, from each technique suggests that ERT is the more precise technique. This observation is confirmed by the $\delta k_{\parallel}$ results shown in the supplementary material.

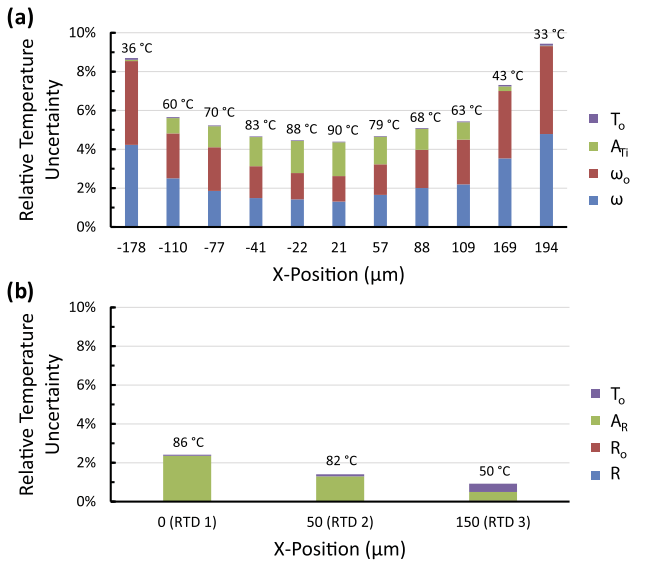


FIG. 8. Relative experimental temperature uncertainties, $\delta T_{exp}/T_{exp}$, propagated according to Eq. (4) from the (a) PaRT and (b) ERT experiments for evaluating $k_{\parallel}$ of sample 2a. The fractional contributions of the component measurement uncertainties (R, R_o, ω, ω_o, etc.) to each relative temperature uncertainty are indicated in the chart legend. The data labels indicate T_{exp} measured from each TiO_2 particle or RTD. The x-position of each TiO_2 particle or RTD was measured with respect to the centerline heater (RTD 1).

C. Sensitivity analysis

Beyond the primary utility of propagating experimental uncertainties, extending MC simulation to include stochastic sampling of FEM input parameter uncertainties (e.g., from other material properties, boundary conditions, etc.) is computationally impractical. The long simulation time typical of finite element analysis (~1 min or greater) would dramatically compound the iterative MC simulation time. Therefore, a more appropriate alternative for approximating the propagated uncertainty contributions of the FEM input parameters is by means of sensitivity analysis.

Generally, sensitivity is an assessment of how strongly a model output result is affected by a small change to an input parameter and can be treated as an indirect measure of uncertainty. For our

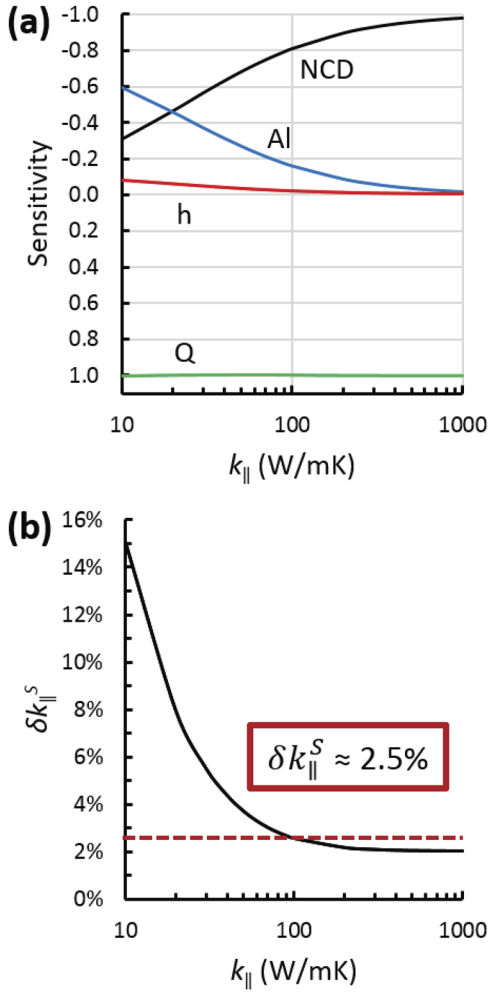


FIG. 9. Sensitivity analysis results of an NCD suspended membrane with backside-deposited Al for which input parameters are listed in Table I. (a) Sensitivity as a function of (NCD) $k_{\parallel}$. $S_{k_{\parallel}}$ and $S_{t_{NCD}}$ are indistinguishable and shown as one curve (black) labeled "NCD." $S_{k_{Al}}$ and $S_{t_{Al}}$ are indistinguishable and shown as one curve (blue) labeled "Al." S_{η}, S_{G_1}, S_{G_2}, S_{G_3}, and S_{T_b} are negligibly low and omitted for clarity. (b) Sensitivity-based uncertainty in $k_{\parallel}$ ($\delta k_{\parallel}^{S}$) as a function of $k_{\parallel}$ demonstrates that $\delta k_{\parallel}^{S} \approx 2.5\%$ for our NCD suspended membrane samples.

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FEM, we defined sensitivity as an assessment of the relative change in the temperature rise resulting from a 5% relative change in an input parameter according to the following equation:

$$S_p = \frac{\delta\Delta T/\Delta T}{\delta p/p} = \frac{\delta\Delta T/\Delta T}{0.05}, \tag{5}$$

where S_p is the sensitivity to input parameter, p; $\delta\Delta T/\Delta T$ is the relative change of output temperature rise, ΔT; and $\delta p/p$ is the relative change of the input parameter, p. Assuming that the auxiliary input parameter uncertainties (not $\delta k_\parallel$) are known and uncorrelated, the unknown fitting parameter uncertainty ($\delta k_\parallel$) can be expressed in terms of the known parameter uncertainties by summation in quadrature[38] according to the following equation:

$$\delta k_\parallel^S = k_\parallel \sqrt{\sum_p \left(\frac{S_p}{S_k}\frac{\delta p_{exp}}{p}\right)^2}, \tag{6}$$

where $\delta k_\parallel^S$ is the sensitivity-based uncertainty in $k_\parallel$ propagated from the experimental input parameter uncertainties, δp_{exp}; and S_p and S_k are the sensitivities to the input parameters, p, and fitting parameter, $k_\parallel$, respectively. Ideally, sensitivity to the fitting parameter should be high ($S_k > 0.2$[19]) and sensitivity to the auxiliary input parameters should be low ($S_p < 0.05$[19]) to obtain a reasonable propagated uncertainty ($\delta k_\parallel^S/k_\parallel < 10\%$). These ideal circumstances permit relatively high auxiliary input parameter uncertainties ($\delta p_{exp}/p \approx 25\%$) without affecting the propagated uncertainty.[19]

The results of our sensitivity analysis of an NCD suspended membrane with backside-deposited aluminum are shown in Fig. 9. The FEM input parameters explored in the sensitivity analysis are listed in Table I. Since S_η, S_{G_1}, S_{G_2}, S_{G_3}, and S_{T_b} were found to be negligibly low across the $k_\parallel$ domain, inaccurate or highly uncertain measurements of these input parameters only negligibly propagated to $\delta k_\parallel^S$. Furthermore, since S_ε was negligibly low and S_h only became non-negligible at low thermal conductivities ($k_\parallel \leq 10$ W/m K), inaccuracies inherent to boundary condition approximations also only negligibly propagated to $\delta k_\parallel^S$. However, the high sensitivity to t_{NCD} indicated that the accurate measurement of t_{NCD} was critical for accurately resolving $k_\parallel$—a common requirement among thin film thermal property measurement techniques.

V. ERROR ANALYSIS

Rigorous uncertainty estimation is an essential component for precise characterization of thin film thermal properties. However, even a perfectly accurate and precise experimental technique can lead to inaccurate thermal property results if the errors inherent to the comparative model are not carefully considered. The model errors relevant to deducing $k_\parallel$ according to the experimental methods presented in this work are those that arise from model assumptions that prevent perfect reproduction of the temperature distribution present in experiment. The model assumptions that meet this criterion were those that involved boundary conditions and material property assignments. Moreover, the only relevant boundary condition or material property assumption not considered via sensitivity analysis was the assumed temperature independence of NCD $k_\parallel$.

Therefore, to validate this assumption, $k_\parallel(T_{avg})$ of samples 1b, 2a, and 2b was measured via ERT according to the methods presented in Sec. III B. An example of these $k_\parallel(T_{avg})$ results is shown in Fig. 10 and demonstrates the minimal temperature dependence that is consistent with experimental $k(T)$ results measured from NCD films reported in the literature.[39] Note that $k_\parallel$ is expressed as a function of average membrane temperature, T_{avg}, since $k_\parallel$ is a membrane-averaged result measured from a gradient of temperatures (see Fig. 12). The elevated $\delta k_\parallel$ values measured for $T_{avg} < 50\,^\circ$C ($\Delta T_{avg} < 20\,^\circ$C) were consistent with similar suspended membrane thermal conductivity uncertainty analysis results presented in the literature[23] that observed excessive δk determined from small

TABLE I. FEM parameters evaluated in the sensitivity analysis shown in Fig. 9 of an NCD suspended membrane with backside-deposited aluminum Al.

Parameter	Symbol	Nominal value	Units	Relative measurement uncertainty (%)
NCD in-plane thermal conductivity	$k_\parallel$	10–1000	W/m K	See Fig. 9
NCD thickness	t_{NCD}	1.0[a]	μm	2[a]
NCD anisotropy ratio ($k_\perp/k_\parallel$)	η	2.8[b]		5
Al thermal conductivity	k_{Al}	200	W/m K	5
Al thickness	t_{Al}	100[a]	nm	6[a]
NCD/Al interface conductance	G_1	147[c]	MW/m^2 K	12[c]
NCD/Si interface conductance	G_2	70[c]	MW/m^2 K	20[c]
RTD/NCD interface conductance	G_3	169[c]	MW/m^2 K	9[c]
Input heating rate	Q	200.0	mW	0.1
Sample base temperature	T_b	30.0	$^\circ$C	20
Convection coefficient	h	N/A	W/m^2 K	5
Thermal emissivity (NCD and RTD)	ε	0.02		5

[a]Measured via transmission electron microscopy (TEM).
[b]Reference 10.
[c]Measured via TDTR.

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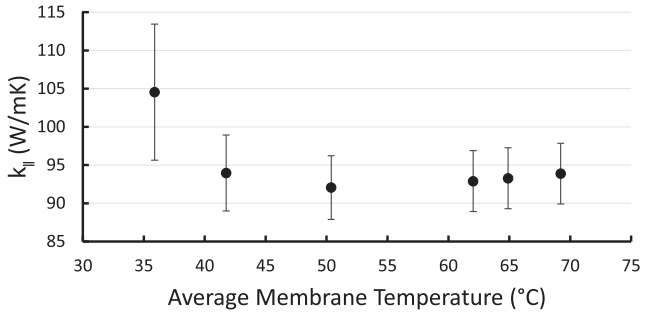


FIG. 10. Experimental $k_{\parallel}$ results as a function of average membrane temperature, T_{avg}, measured from sample 1b via ERT at various power conditions. Large $k_{\parallel}$ and $\delta k_{\parallel}$ for $T_{avg} < 50\,^{\circ}\text{C}$ were attributed to ERT-specific errors incurred at low average temperature rise, $\Delta T_{avg} = T_{avg} - T_{min}$, where T_{avg} and T_{min} are the average and minimum suspended membrane temperatures, respectively.

temperature rises ($\Delta T < 20$ K). Since our experimental $k_{\parallel}$ results shown in Fig. 12 demonstrate that the PaRT technique is capable of accurately resolving $k_{\parallel}$ for low temperature rises ($\Delta T = T_{max} - T_{min} = 22\,^{\circ}\text{C}$), it is clear that the elevated $k_{\parallel}$ and $\delta k_{\parallel}$ measured for $T_{avg} < 50\,^{\circ}\text{C}$ via ERT are a consequence of the ERT technique as opposed to the NCD material. Therefore, we attribute the elevated $k_{\parallel}$ and $\delta k_{\parallel}$ measured for $T_{avg} < 50\,^{\circ}\text{C}$ via ERT to the measurement or modeling error inherent to our implementation of the ERT technique. Methods for mitigating this error have been discussed in Sec. V.

In addition to enabling a rigorous assessment of model error, utilizing a FEM as the comparative model for deducing $k_{\parallel}$ was particularly useful for accommodating the extraneous back-side aluminum thin film present on our suspended membrane samples. However, in the absence of extraneous thin film coatings, it is possible to drastically simplify post-measurement analysis by assuming 1D thermal conduction across the entire suspended membrane and applying Fourier's law according to the following equation:

$$k_{\parallel}^{1D} = -\frac{Q}{tL}\left(\frac{\Delta x}{\Delta T}\right), \tag{7}$$

where $k_{\parallel}^{1D}$ is the suspended membrane $k_{\parallel}$ assuming 1D conduction, Q is half the heating rate sourced from RTD 1, t is the suspended membrane thickness, L is the length of RTD 1, ΔT is the temperature rise ($T_{max} - T_{min}$) sampled from the surface of the 1D conduction regime, and Δx is the lateral distance between T_{max} and T_{min}.

The modeling error resulting from this 1D conduction assumption, ε_{1D}, can be assessed independently of $\delta k_{\parallel}$ by performing a comparative analysis between the FEM and Fourier law according to the following equation:

$$\varepsilon_{1D} = \frac{k_{\parallel}^{1D} - k_{\parallel}^{FEM}}{k_{\parallel}^{FEM}}, \tag{8}$$

where $k_{\parallel}^{1D}$ is evaluated according to Eq. (7) with ΔT sampled from FEM results and $k_{\parallel}^{FEM}$ is the $k_{\parallel}$ input to the FEM. The results of this comparative analysis are presented in Fig. 11 and demonstrate that deducing $k_{\parallel}$ according to a simple 1D approximation results in $\varepsilon_{1D} < 5\%$ for $k_{\parallel} > 90$ W/m K and for membrane dimensions with a length-to-width aspect ratio greater than 4.5. FEM results (not shown) have demonstrated that NCD suspended membranes with $t > 5$ μm deviate from the 1D condition by developing a temperature gradient through the thickness of the suspended membrane. In the absence of extraneous thin film coatings, $k_{\parallel}^{1D}$ can be experimentally approximated by evaluating Eq. (7) with ΔT and Δx sampled from the inner- and outermost particles on the suspended membrane surface via PaRT or from RTDs 2 and 3 via ERT. Hence, $\delta k_{\parallel}^{1D}$ would be evaluated according to the analytical derivation uncertainty propagation method (see Sec. IV B).

VI. EXPERIMENTAL RESULTS

The experimental $k_{\parallel}$ results measured from the NCD suspended membrane samples via PaRT and ERT are presented in Table II. The experimental temperature profiles measured via PaRT and ERT and the histograms resulting from fitting $k_{\parallel}$ for sample 2a are shown in Fig. 12. The precise agreement between $k_{\parallel}^{PaRT}$ and $k_{\parallel}^{ERT}$ measured from sample 2a demonstrates the accuracy of the two independent steady-state techniques. In addition, both PaRT and ERT have been demonstrated capable of resolving $k_{\parallel}$ of the anisotropic NCD thin films with $\delta k_{\parallel} \leq 5\%$, a membrane-averaged result that is a substantial improvement upon the $\delta k_{\parallel} \approx 11\%$–$15\%$ resolved from TDTR measurements of similar NCD thin films.[13] Furthermore, TDTR measurements are inherently sensitive to spatial material property variability that can add considerable uncertainty to spatially averaged $k_{\parallel}$ result. For example, a recent demonstration of 2D thermal conductivity mapping via spatially resolved TDTR[9] reported that up to 35 independent TDTR

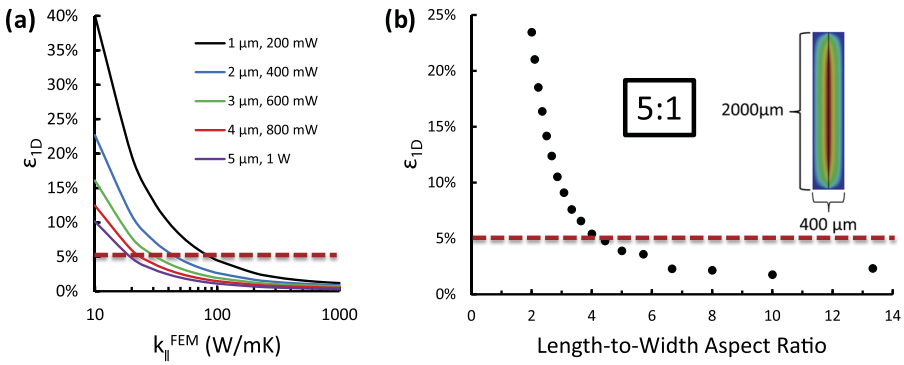


FIG. 11. Error-based criteria for deducing $k_{\parallel}$ according to the simplifying assumption of 1D thermal conduction resulting from comparative model error analysis between the FEM and Fourier law assuming 1D conduction. The dashed line indicates the conditions that yield $\varepsilon_{1D} < 5\%$. (a) ε_{1D} as a function of $k_{\parallel}^{FEM}$. Linear power density along RTD 1 [Q/t in Eq. (7)] was held constant to maintain a nearly constant ΔT. (b) ε_{1D} as a function of length-to-width aspect ratio.

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TABLE II. Experimental $k_{\parallel}$ results obtained from PaRT, ERT, and the 1D conduction approximation.

Sample	NCD film thickness (μm)	Al film thickness (nm)	Nucleation seed size (nm)	$k_{\parallel}^{PaRT}$ (W/m K)	$k_{\parallel}^{ERT}$ (W/m K)	$k_{\parallel}^{1D}$[a] (W/m K)
1a	1.06 ± 0.02	92 ± 5	4	99 ± 4 (4%)	. . .[b]	N/A[c]
1b	1.06 ± 0.02	92 ± 5		. . .[b]	94 ± 4 (4%)	N/A[c]
2a	1.06 ± 0.05	100 ± 5	20	117 ± 5 (4%)	118 ± 4 (3%)	N/A[c]
2b	1.06 ± 0.05	100 ± 5		. . .[b]	118 ± 4 (3%)	N/A[c]
3	1.01 ± 0.05	0[d]	. . .[b]	102 ± 5 (5%)	. . .[b]	103 ± 9 (8%)
4	3.02 ± 0.15	100 ± 5	. . .[b]	157 ± 8 (5%)	. . .[b]	N/A[c]

[a] $k_{\parallel}^{1D}$ was experimentally approximated via PaRT by evaluating Eq. (7) with ΔT and Δx sampled from the inner- and outermost TiO_2 particles on either half of the suspended membrane.
[b] Data were not measured.
[c] 1D approximation was not valid due to the presence of backside Al on the suspended membrane.
[d] No backside Al present.

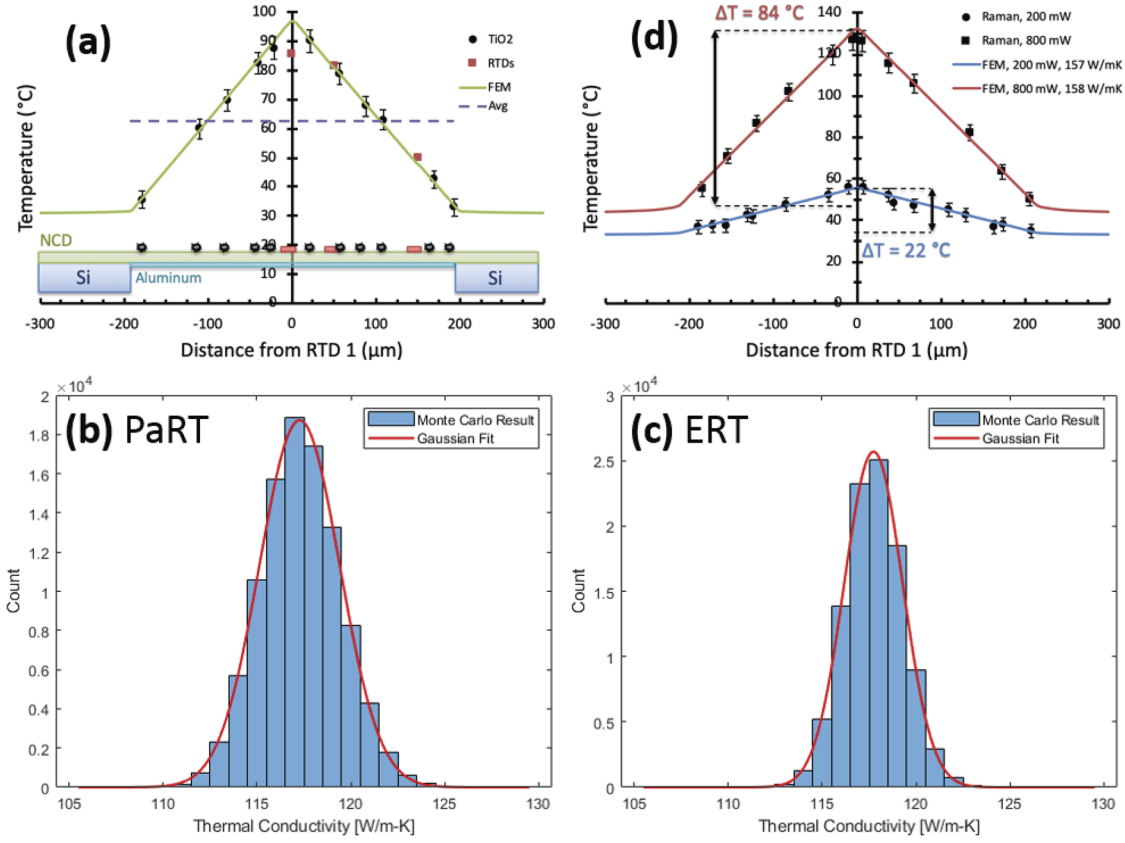


FIG. 12. (a) Temperature profile measured from sample 2a via PaRT and ERT with 200 mW dissipated from RTD 1. Both steady-state thermometry techniques demonstrate precise agreement with the FEM. Note that the temperature measured from RTD 1 is lower than the linear trend predicted by the FEM. This result was expected and is explained in Sec. III B. The average membrane temperature, T_{avg} = 63 °C, calculated via the FEM is denoted by the dashed line. (b) PaRT and (c) ERT Monte Carlo simulation histograms for determining $k_{\parallel}$ and $\delta k_{\parallel}^{MC}$ for sample 2a. (d) Temperature profile measured from sample 4 (3 μm) via PaRT with 200 and 800 mW dissipated from RTD 1. The $k_{\parallel}$ results of 157 ± 8 and 158 ± 8 W/m K measured with temperature rises of 22 and 84 °C, respectively, demonstrate the possibility of resolving $k_{\parallel}$ for low ΔT via PaRT.

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measurements would be necessary to achieve a spatially averaged result that lies within ±10% of the global mean with a probability >95% (for a relative spatial variability of 30%), a proposition that entails prohibitively long measurement times using standard TDTR mapping implementations. Moreover, leveraging the simplifying assumption of 1D thermal conduction according to the error-based criteria shown in Fig. 11 permits drastically reduced post-measurement analysis for deducing $k_{\parallel}$ with minimal sacrifice to accuracy and precision. The $k_{\parallel}$ results measured from sample 3 (no backside Al) shown in Table II demonstrate the experimental applicability of this simplifying 1D conduction approximation. The superior accuracy and precision demonstrated by the PaRT and ERT techniques provide clarity to distinguish $k_{\parallel}$ differences as small as 20 W/m K between NCD thin films fabricated according to different growth conditions, as shown in Table II. Such comparative analyses have been integral to improving the material quality and thermal properties of NCD thin films for heat spreading applications such as GaN-on-diamond HEMT device technology.[40]

VII. CONCLUSIONS

In order to demonstrate the efficacy, practical limitations, and best practices for measuring $k_{\parallel}$ of high thermal conductivity thin film materials via steady-state thermometry techniques, we have developed an explicit framework for rigorously considering experimental uncertainty and model error. This uncertainty and error analysis framework has been applied to two independent steady-state thermometry techniques: PaRT and ERT. Furthermore, we have established error-based criteria to determine the limiting experimental conditions that permit the simplifying assumption of 1D thermal conduction to drastically reduce the post-measurement analysis necessary for deducing $k_{\parallel}$. The superior accuracy and precision ($\delta k_{\parallel} < 5\%$) of the steady-state PaRT and ERT techniques have been demonstrated by measuring highly disordered and anisotropic NCD suspended thin films. Our experimental results have demonstrated that the PaRT technique is capable of accurately resolving $k_{\parallel}$ for low temperature rises ($\Delta T = T_{max} - T_{min} = 22\,^{\circ}\mathrm{C}$) and that the ERT technique is capable of quickly measuring an approximation of the suspended membrane temperature-dependent thermal conductivity, $k_{\parallel}(T_{avg})$. Although our experimental $k_{\parallel}(T_{avg})$ results have confirmed the assumed temperature independence of NCD $k_{\parallel}$, this assumption is not required since $k_{\parallel}(T)$ can easily be incorporated into the numerical finite element model (FEM). We have established error-based criteria for deducing $k_{\parallel}$ with less than 5% error according to the simplifying assumption of 1D thermal conduction resulting from comparative model error analysis between the FEM and Fourier law assuming 1D conduction. The accuracy of exercising this 1D approximation experimentally has been demonstrated by the precise agreement of $k_{\parallel}$ results measured from the same sample with and without the simplifying assumption. Furthermore, the strength of using a numerical FEM for model fitting has been demonstrated by resolving accurate NCD $k_{\parallel}$ in the presence of a backside Al thin film, a capability not easily accomplished when using an analytical model.

In summary, the superior accuracy and precision ($\delta k_{\parallel} < 5\%$) of the steady-state PaRT and ERT techniques have been demonstrated to be ideal methods for characterizing the in-plane thermal conductivity, $k_{\parallel}$, of high thermal conductivity ($k_{\parallel} > 50$ W/m K) thin films ($t < 5\ \mu$m) for heat spreading applications. Additionally, the error-based criteria established for assuming 1D conduction provide guidelines for suspended membrane design in future studies of other high thermal conductivity materials using these steady-state PaRT and ERT techniques.

SUPPLEMENTARY MATERIAL

See the supplementary material for details about the FEM, temperature sensor calibrations, and component measurement uncertainties.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The data that support the findings of this study are available within the article and its supplementary material.

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